

The Art of Cloth simulation on The Boss Baby : Family Business

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Figure 1: From Downsizing characters (a) to Animated scaling computation (b) to a massive character catalog (c)

ABSTRACT

DreamWorks Animation's *The Boss Baby : Family Business* has a strong design language. Clothing is an essential part of characterization we needed to maintain strong graphic silhouettes on garments. The characters love changing clothes pretty frequently. Additional Challenges included dream sequences, animated scaling and downsizing of characters, with an extensive catalog and large crowds wearing multiple layered winter clothes. This talk illustrates various techniques used to tackle those and the use of animation-driven physics based garment simulation to emphasize the art of dressing an animated feature.

CCS CONCEPTS

• **Computing methodologies** → **Physical simulation; Shape modeling.**

KEYWORDS

cloth, simulation, art-driven, shapes, sculpting, scaling, variants, scaling, downsizing

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1 DESIGN COMPLEXITY OF CLOTHING

The Boss Baby : Family Business has two visually contrasting styles of animation; certain characters have a tightly fitted, organized look with methodical performances, while others are messy and disoriented with snappy animation. To define the appearance and mannerism of these characters, clothing is an essential element of the character design and performance. Cloth and hair are pushed in Character Effects, most of which is added post animation, enhancing the contrasts in the characters. Messy characters go through several asymmetry passes and various levels of dishevelment for each sequence in their cloth simulation. Additionally, just like the real world, our primary characters love changing clothes frequently. Since the movie is set in winter, most of the characters have over 20 different multi-layered costumes, as illustrated in Fig. 3. The movie also features heavily art-directed graphical fantasy sequences with a variety of costumes for one-off instances, which required shot specific garment design and simulation.

2 ART-DRIVEN CLOTH SIMULATION

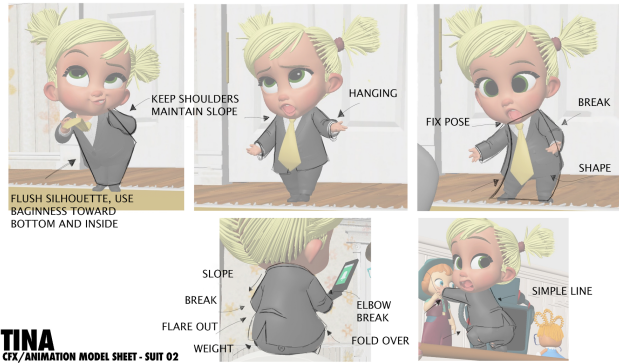


Figure 2: Tina CFX Model sheet, describing the design language

A strong graphic silhouette defines the characters. The Animation team adopted a snappy classic 2D animation style with smear effects on geometries instead of motion blur which demanded a lot of follow-through and overlap on clothes. The CFX team worked with the Rigging department to provide Animation with additional controls on the cloth to articulate the graphic shaping of garments. CFX designed a workflow around it, to preserve the original shapes in the cloth simulation. Animation shapes were used as an input for the simulations by painting goal shapes to preserve and enhance the silhouettes. Character sheets included libraries of drawings for cloth as a series of rules to stay on model and maintain the visual style as shown in Fig. 2. Additionally, characters wear sweaters that mimic real world knits. It was essential to preserve the volume and structure, counter the stretch and squash from the character performance, add dynamic and naturalistic wrinkles through simulation while maintaining the graphic silhouette. This was achieved through a specific double-layered volumetric mesh construction approach with pressure attributes. The inner layer of volumetric geometry follows the animation driven shapes while the outer layer is a result of pure cloth simulation which makes it look and feel stylized yet real.

The art direction on cloth also led to the development of a camera based shot sculpting tool to graphically shape cloth and hair further. This toolset was also used to draw and keyframe custom wrinkles on geometries post simulation.

3 SCALING AND DOWNSIZING

The movie provided a unique opportunity of simulating cloth over animated scaling characters. In Fig. 1b. Tim is animated scaling down to 65% of his normal size whereas Boss Baby scaled up to 10 times in a single shot. Cloth rigs were specifically developed to leverage the proprietary stretch rig system which dynamically adds or reduces extra fabric. The cloth rig also computed properties of cloth such as mass, stiffness, bend, based on the animated scale of the character for a seamless simulation while maintaining the look of the garment. Along with cloth, similar effects were also added on the hair which would dynamically update the hair styling attributes such as width and kink based on scaling.

One of the most challenging aspects of the movie was a transformation sequence which we referred to as Downsizing in Fig. 1a. Boss Baby and Tim Adult transform into their younger versions, gradually morphing on camera and going through different stages of childhood. While the characters were shrinking inside their oversized costumes, cloth and hair simulation enhanced the performance of this whole act. The Animation team came up with performances of characters without the clothing and mostly relied on drawings to convey the idea, based on which the simulation was executed. This was a collaborative process between the Animation and CFX teams to choreograph the Downsizing sequence and bring it to life.



Figure 3: Variety of costumes for each character

4 VARIETY AND DIVERSITY

With over 900 characters with multi-layered costumes dressed for the winter, the primary characters carry a large wardrobe with up to 20 costumes each. The school sequences in the movie shown in Fig. 1c, boasted over 400 types of kids with different body types and a huge catalog of clothes to pick from. We revamped our procedural cloth builder tool and developed a new multi-variant cloth rig workflow to handle a large volume of clothing. This allowed us to pick and choose from the hundreds of different costume combinations available for the generic crowd characters. The cloth rigs were procedurally built into master simulation files containing all the garments available and allowing all possible permutations and combinations. During shot work, the multi-variant system could automatically filter out and install the relevant cloth components needed for each character or crowd-ant respectively. This system helped us to maintain a minimum amount of cloth rigs files.

5 CONCLUSION

This agile cloth workflow developed on the movie alleviated underlying structural complexity and allowed the artists to focus on the many artistic challenges. It enabled more iterations quicker to push the style to new levels.

The innovative clothing solutions adopted on The Boss Baby : Family Business, enabled our modest CFX team to tackle more challenges such as simulating entire bedding sequences, sleeping bags, blankets, curtains, ropes, seatbelts, backpacks, carpets, table cloth, christmas trees, foliage, wind to zero gravity and from underwater effects to wet looks.